

Transmission Line Faults Classification Based on Alienation Coefficients of Current and Voltage Waveform and SVM

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Abstract—In order to further improve the speed and accuracy of transmission line fault classification, a new method based on alienation coefficients of the combination of fault current and voltage waveform is proposed in this paper, which uses the support vector machine(SVM) as the classification tool. This method analyzes the current and voltage waveforms simultaneously of a quarter cycle after a fault occurs, and calculates the waveform alienation coefficients on the total 8 signal channels of the three-phase and neutral lines of the current and voltage respectively, then composes them as input features vector, finally completes the fault classification by using SVM model. The experiments compare the classification effect when different signal types are used, and test the robustness and applicability of the model by changing the sampling rate and adding noise interference. Simulation results show that the method can quickly classify faults after transmission line faults occur with a classification accuracy rate reaches 99.94%, which has great practical application potential.

Keywords—transmission line, fault classification, waveform, alienation coefficient, SVM

I. INTRODUCTION

With the development of modern science and technology, people have higher and higher requirements for power quality and reliability. If a large-scale system blackout occurs, it will cause unpredictable harm to all aspects of human life. Due to the wide coverage of transmission lines and the harsh operating environment, about two-thirds of total failures caused by transmission line fault. Each line trip caused by the transmission line fault will have a huge impact on the power system. Once handled improperly, it will seriously affect the safe and stable operation of the system[1]-[3]. Therefore, fast and accurate fault classification can greatly improve the reliability of power supply.

In the present study, the process of fault classification of transmission lines is usually divided into two steps: the first step is to extract fault features; the second step is to use artificial intelligence models or compare the obtained feature coefficients with preset thresholds to achieve fault classification.

Reference [4] utilized multi-resolution analysis(MRA) in discrete wavelet transform(DWT) to decompose the fault signal, extracted the detailed coefficients containing its fault information, and combined k-Nearest Neighbor(KNN) to achieve the transmission line fault classification. Reference [5][6] adopted DWT to decompose the fault signal, and it used the wavelet energy obtained from decomposition as the fault feature to input to the artificial neural network (ANN) to achieve classification of fault types. Reference [7] defined four types of multi-wavelet entropy, including energy entropy, time entropy, Shannon singular entropy, and Tsallis singular entropy, to extract the feature of transmission line faults, and adopted RBF neural network as the fault type classifier. The above methods all have relatively high classification accuracy. However, their model requires half-cycle or even one-cycle fault recording data for accurate classification after the fault occurs, which takes a long time.

Another type of method commonly used in transmission line protection is to compare the feature coefficients during the occurrence of the fault with a preset threshold. Reference [8] used Db4 as the mother wavelet of wavelet transform, and decomposed the fault signal to extract wavelet singular entropy (WSE). By comparison, it found that the WSE of the fault phase is higher than the normal phase. In [9], DWT was used to extract the feature of the fault current signal, and then a wavelet matrix was established, and the maximum wavelet singular value of the matrix was obtained by using singular value decomposition(SVD). The method proposed in [10] was to calculate the alienation coefficient of the current waveforms on

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the three-phase and neutral lines before and after the occurrence of the fault, and took alienation coefficient as an index to determine the type of the fault. The above methods are to compare the obtained feature value with the preset threshold value to achieve fault classification. However, it is a challenge to select the appropriate threshold to satisfy various operating environments.

In summary, this paper proposes a new method which is based on the alienation coefficients of both fault current and voltage waveform, and uses SVM to classify the transmission line fault. Its main advantages are as follows: firstly, the method is able to effectively extract the time-domain feature of fault current and voltage, and achieve high classification accuracy; secondly, the calculation process is simple, and only a quarter cycle of fault data is required, which improves the classification speed.

II. ALGORITHM AND THEORETICAL BASIS

A. Alienation coefficient

When a fault occurs on the transmission line, the current and voltage waveforms of the faulty phase will be greatly different from the normal ones, and the changes of the non-faulty phase waveform will be relatively small [11]. Figure 1 shows how the waveform changes during a transmission line fault. In this paper, the voltage or current waveform signals $x(n)$ of a quarter cycle after the fault on the three-phase and neutral lines are selected, and the degree of alienation is calculated relative to the normal waveform signals $y(n)$ of the same polarity before the fault. Its expression is as follows:

$$A = (1 - \rho_{xy}^2) \quad (1)$$

$$\rho_{xy} = \sum_{n=0}^{N-1} x(n)y(n) \left(\sum_{n=0}^{N-1} x^2(n) \sum_{n=0}^{N-1} y^2(n) \right)^{-\frac{1}{2}} \quad (2)$$

where:

A is the alienation coefficient, which reflects the degree of alienation between the two sets of waveforms;

ρ_{xy} is the correlation coefficient between the waveform signals $x(n)$ and $y(n)$;

$x(n)$ is waveform data of a quarter of a period after the time t_0 when the fault occurred;

$y(n)$ is the normal waveform sample of the previous fault cycle and the first quarter cycle after $(-T + t_0)$ time;

N is the number of data points sampled in the quarter cycle waveform (in this article $N = 50$).

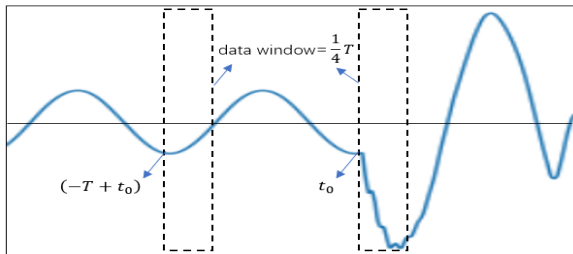


Fig. 1 Data window selection

B. Basic principle of SVM

SVM is a machine learning algorithm based on structural risk minimization. When the size of samples is small, it can effectively avoid over fitting phenomenon while maintaining strong generalization ability and classification effect [12][13]. The network structure of SVM is shown in Figure 2 and the main idea of its classification is as follows:

- When the samples are linearly separable in the sample space, there is a hyperplane which is farthest from the two types of samples. This hyperplane should be robust enough to tolerate local disturbances in training samples. Let the expression for dividing the hyperplane be:

$$f(x) = w^T x + b \quad (3)$$

Where:

w is the normal vector that determines the direction of the hyperplane;

b determines the distance between the hyperplane and the origin.

- In the case where the samples are non-linearly separable, such a suitable hyperplane may not be found in the original sample space. At this time, by increasing the original sample space to a higher dimension, the samples can be linearly separable after mapping to the new feature space. Suppose the expression used to divide the hyperplane at this time is as follows:

$$f(x) = \sum_{i=1}^m \alpha_i \gamma_i \kappa(x, x_i) + b \quad (4)$$

Where:

m is the number of support vectors;

$\kappa(x, x_i)$ is the kernel function.

- Whether the sample can be linearly separable in the new feature space, the choice of the kernel function plays a key role, because any kernel function implicitly defines a feature space. Table I lists the expressions of four commonly used kernel functions. However, for specific engineering problems, there is still no uniform method for selecting the kernel function. According to the statistical experience, this article belongs to the case where the number of features is small and the number of samples is not large, so the RBF kernel function should be tried first [14].

TABLE I. COMMON KERNEL FUNCTIONS

Kernel	Formula
Linear	$\kappa(x_i, x_j) = x_i^T x_j$
Poly	$\kappa(x_i, x_j) = (x_i^T x_j)^d, d \geq 1$
RBF	$\kappa(x_i, x_j) = \exp\left(-\frac{\ x_i - x_j\ }{\sigma}\right), \sigma > 0$
Sigmoid	$\kappa(x_i, x_j) = \tanh(\beta x_i^T x_j + \theta), \beta > 0, \theta < 0$

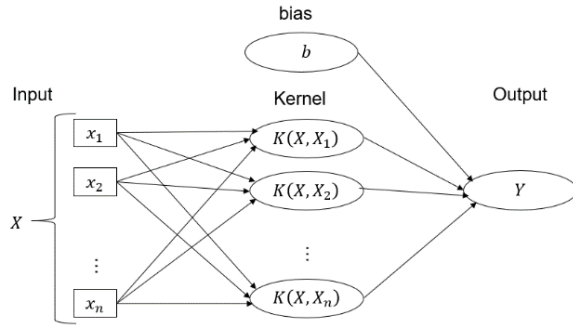


Fig. 2 Network structure of support vector machine

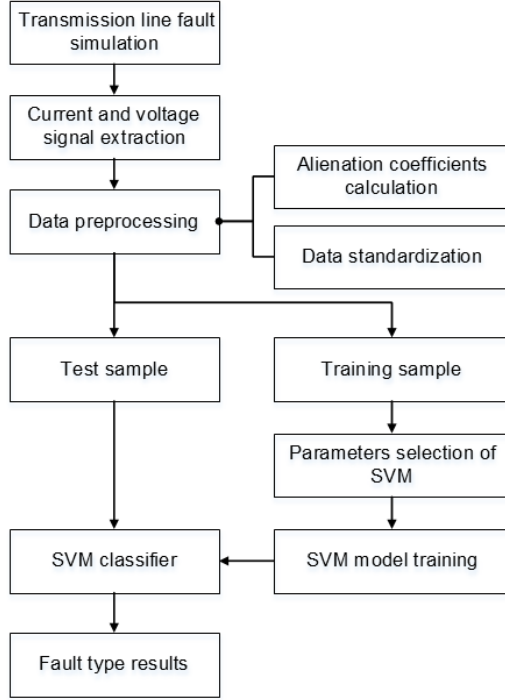


Fig. 3 Fault classification model flowchart

III. EXPERIMENTAL PROCESS

The flowchart of transmission line fault classification experiment in this paper is shown in Figure 3. The specific steps of the experiment are shown below:

- Step 1 Use PSCAD/EMTDC to model and simulate transmission line faults, and collect current and voltage signal data on three-phase and neutral lines;
- Step 2 Calculate the alienation coefficients on 8 data channels and normalize them;
- Step 3 Sequentially construct the normalized alienation coefficients into the input feature vector $P = [A_{I_a}, A_{I_b}, A_{I_c}, A_{I_0}, A_{U_a}, A_{U_b}, A_{U_c}, A_{U_0}]^T$ and randomly divide the feature vector samples into a training set and a test set according to a ratio of 7-3;
- Step 4 Select appropriate parameters and train the SVM model;
- Step 5 Use the trained SVM model to predict the test sample data.

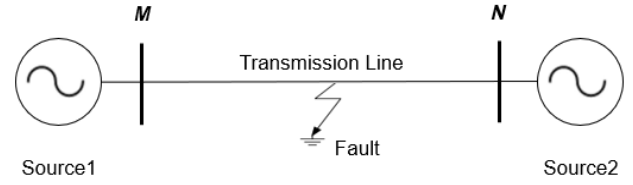


Fig. 4 Simplified model of transmission line

TABLE II. FAULT PARAMETERS SETTING

System Parameter	Values and Types
Fault Distance(km)	20, 60, 100, 140, 180
Fault Resistance(Ω)	0~100
Fault Inception Angles(degrees)	0, 30, 60, 90
Fault Type	A-g, B-g, C-g, AB, AC, BC, AB-g, AC-g, BC-g, ABC-g

A. Transmission line simulation model

A fault simulation system for a high-voltage transmission line with a voltage level of 220kV and a length of 200km was established in PSCAD/EMTDC, as shown in the figure 4. The line model in the system uses Frequency Dependent (Phase) Model. The relevant parameters of the transmission line were set as follows:

Positive sequence parameters:

$$R_1 = 3.648 \times 10^{-2} \Omega / km, L_1 = 1.348 mH / km,$$

$$C_1 = 8.68 \times 10^{-3} \mu F / km$$

Zero sequence parameters:

$$R_0 = 0.3 \Omega / km, L_0 = 3.639 mH / km,$$

$$C_0 = 6.166 \times 10^{-3} \mu F / km$$

$$\text{System frequency : } f_s = 50 Hz$$

$$\text{Sampling frequency : } f_c = 10 kHz$$

B. Data Acquisition

In this paper, by using the simulation model, a large amount of fault recording data can be obtained, thereby making up for the problem of insufficient fault data of real transmission lines. The setting of fault parameters are listed in table II. By using Multiple Run components in PSCAD, multiple simulations can be conveniently performed at the same time, which greatly improves the simulation efficiency during fault simulation. In the process of simulation, we performed 600 different combinations of fault parameters for each type of fault, and collected current and voltage recording data on the M-terminal bus at the same time. Therefore, we obtained $600 \times 10 = 6000$ sets of fault sample data.

C. Data preprocessing

1) Calculation of Alienation Coefficient

Select the data of a quarter cycle after the transmission line fault occurrence time t_0 and the data of a quarter cycle with the same polarity as the normal waveform of the cycle before the fault. Since the system frequency is 50Hz and the sampling frequency is 10kHz, there are 200 sampling points in one cycle,

corresponding 50 sampling points in a quarter of the cycle. Combining equations (1) and (2), the waveform alienation coefficient A is calculated.

2) Data normalization

The normalization processing is performed on the waveform alienation coefficient A , that is, the obtained waveform alienation coefficient A is linearly scaled in $[0,1]$, which is beneficial to the training of the SVM model. The normalization formula is as follows:

$$A_{norm} = \frac{A - A_{min}}{A_{max} - A_{min}} \quad (5)$$

Where:

A 、 A_{norm} are the values before and after normalization;

A_{max} is the maximum value in A ;

A_{min} is the minimum value in A .

3) Construct the input feature vector

For each sample, the voltage and current recording data of its three phases and neutral lines are obtained, so there are 8 sets of fault recording data. Then construct the normalized 8 alienation coefficients into the input feature vector:

$$P = [A_{I_a}, A_{I_b}, A_{I_c}, A_{I_0}, A_{U_a}, A_{U_b}, A_{U_c}, A_{U_0}]^T$$

4) Setting of fault sample label

In this paper, 10 types of transmission line faults A-g, B-g, C-g, AB, AC, BC, AB-g, AC-g, BC-g, and ABC-g are labeled from 0 to 9.

D. Optimization of SVM parameters

In this paper, the SVM model utilizes radial basis function(RBF), so there are only two parameters c and g that need to be optimized in the model. We randomly select 4200 sets of training samples from 6000 fault samples, and use StratifiedKFold's method to optimize parameters. The specific implementation steps of this method are as follows:

1) Set the range of parameters $c \in [2^{-10}, 2^{10}]$, $g \in [2^{-10}, 2^{10}]$

2) This paper sets $K=5$, that is, the sample is evenly divided into 5 groups according to a certain rule. For each prediction, 4 sets of data are selected as the training set, and the remaining 1 set is used as the test set. The accuracy of each prediction is recorded through cross validation.

3) A set of c and g values with the highest prediction accuracy are selected as the best parameters.

Finally, the optimal parameters apply in this paper are $c=1, g=16$.

IV. EXPERIMENTAL RESULTS

The remaining 1800 sets of fault sample data and labels are used as the test sets to verify the trained SVM model.

A. Fault classification performance evaluation

In Table III, the classification accuracy and overall classification accuracy of the 10 fault types are recorded. The results show that the method based on fault waveform alienation coefficient and SVM classification has obvious advantages on

fault classification with the overall accuracy up to 99.94%. Among them, the classification accuracy of single-phase ground faults with the highest probability of fault occurrence in transmission lines and three-phase short-circuit faults with the most severe impact on system stability reached 100%.

B. Anti-noise performance evaluation

In this paper, Gaussian white noise with different intensities is added to the sample data to test the anti-noise performance of this model. The results are recorded in Table IV. The results show that the accuracy of classification decreases slightly as the noise intensity increases. When the noise intensity is low the classification accuracy remains almost unchanged. Even when the signal-to-noise ratio (SNR) reaches 20dB, the classification accuracy can be maintained above 99.7%, which indicates that the model has good anti-noise performance.

TABLE III. STATISTICS OF FAULT CLASSIFICATION ACCURACY RESULTS

Fault type	Number of samples	Number of mistakes	Classification accuracy(%)
A-g	171	0	100.00
B-g	184	0	100.00
C-g	190	0	100.00
AB	186	0	100.00
AC	174	0	100.00
BC	191	0	100.00
AB-g	193	1	99.48
AC-g	167	0	100.00
BC-g	171	0	100.00
ABC-g	173	0	100.00
Totality	1800	1	99.94

TABLE IV. FAULT CLASSIFICATION ACCURACY AT DIFFERENT NOISE LEVELS

SNR(dB)	Accuracy (%)
No noise	99.94
40	99.94
30	99.89
20	99.72

TABLE V. FAULT CLASSIFICATION ACCURACY UNDER DIFFERENT KERNEL FUNCTIONS

Kernel function	Accuracy(%)
Linear	98.72
RBF	99.94
Poly	98.44
Sigmoid	92.67

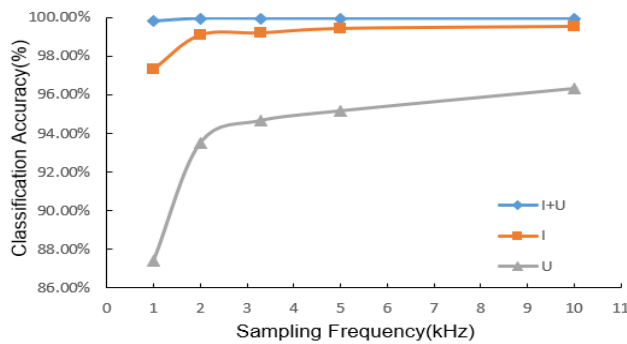


Fig. 5 Comparison of classification accuracy using different signal combinations and different sampling frequencies

C. Comparison of using different kernel functions

Table 1 records the influence of different common used kernel functions on the transmission line fault classification. The results show that the classification accuracy obtained by using RBF as the kernel function is higher than the other three kernel functions in this experiment.

D. Comparison of different recording signal types and sampling frequencies

The above experiments in this paper use data collected at a sampling frequency of 10 kHz. However, the sampling frequency of many fault recorders currently used in the substation is lower than 10 kHz. Therefore, this article also tried to use 1 kHz, 2 kHz, 3.3 kHz, 5 kHz sampling frequency and sample I+U, I and U channels respectively. The results of fault classification accuracy are recorded in Figure 5.

It can be seen from figure 5 that the fault classification effect is best when using the I+U recording signal. Using current signal alone also has a good classification effect, while the classification effect of using voltage signal alone is not ideal. When the sampling frequency is reduced from 10 kHz to 1 kHz, the classification accuracy of I+U recording signal model is reduced from 99.94% to 99.89%, with a decrease of less than 0.1%; when I and U signals are used separately, their classification accuracy is reduced from 99.56% to 97.33%, and 96.33% to 87.39%, with a decrease of about 2% and 9%. This phenomenon is in line with expectations. As the sampling frequency decreases, the fault transient information contained in the recording signal is correspondingly lost, which will affect the classification accuracy of the fault.

V. CONCLUSION

In this paper, a new method for classification of transmission lines fault is proposed. This method analyzes the degree of alienation of fault current and voltage recording signals before and after a fault occurs, and utilizes SVM model as classifier. By establishing a simulation model and carrying out experiments, it is proved that this method can extract fault waveform features quickly and effectively, and only a quarter cycle data is needed before and after fault to achieve fault classification. The final experiment results show that the classification accuracy reaches 99.94%. The proposed method has been tested successfully for various types of faults with the variations in location, inception

angle and fault impedance, and shows good robustness and adaptability to sampling rate changes and Gaussian white noise.

Because the method of transmission line fault classification proposed in this paper is based on experiments with simulation data, how to extend it to engineering application needs further discussion. Finding some real fault data in the power grid to verify the performance of the model, and adding new fault data to the fault database to continuously update and improve the model are the issues that need to be studied in the future.

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